

Economic Effect of Large Sporting Events

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ABSTRACT

Large-scale international events such as the Olympic Games and the FIFA World Cup are frequently cited as catalysts for economic growth and urban development. This paper uses data science methodologies to show the economic growth and urban development in host countries' Gross Domestic Product (GDP) from 1964 to 2020. Through our analysis of historical event data, including costs, revenues, and economic indicators, we developed predictive models using Linear Regression and Random Forest algorithms to assess financial patterns and forecast future trends. Our analysis reveals a persistent and widening gap between event costs and revenues, with costs escalating from an average of 6.0 billion USD in 2032 projections to 6.6 billion USD by 2040, while revenues remain stagnant at approximately 1.5-1.8 billion USD. The Random Forest Regressor achieved an R score of 0.2234 for revenue prediction, while Linear Regression demonstrated superior performance for cost estimation with an R of 0.1087. These findings indicate increasingly negative financial outlooks for future mega-events, with projected losses reaching 5.1 billion USD by 2040, suggesting an urgent need for structural reforms in event planning and funding mechanisms.

BACKGROUND

Major events are often justified on the premise that they will create a lasting economic impact that would be beneficial to the overall economic landscape. Prior research highlights several advantages that would be expected, including: increased Gross Domestic Product (GDP) due to growth in tourists, hospitality, service sectors, big foreign investments, expansion of infrastructure, and creation of employment opportunities.

Despite these projections, studies often show mixed outcome. Many cities face challenges like overspending, underutilization, short-term economic boosts that fail to show long-term and overestimation in growth.

Research on host-related sporting events shows that while some countries experience modest GDP increases,

long-term economic gains are less consistent. Our study applies modern exploratory and predictive techniques to a dataset that covers major sporting events.

RESEARCH QUESTION

The main research question guiding us is, do major international sporting events, specifically the Olympic Games and FIFA World Cup, lead to measurable positive economic outcomes for host cities and countries when evaluated through the lens of cost versus revenues, and what financial patterns emerge from historical data that can inform predictions about future events?

This paper focuses on understanding the direct financial relationship between event cost and revenue generation, examining whether the substantial investments required for hosting these mega-events yield comparable economic returns. The analysis evaluates total costs encompassing venue construction and organizational expenses against total revenues derived from ticketing, broadcasting rights, and commercial sponsorships. This direct cost-revenue comparison provides a clearer picture of financial sustainability than broader GDP impact studies that must account for numerous confounding variables.

The scope of this study encompasses Olympic Games and FIFA World Cup events from 1964 to 2020, providing a 56-year-long-term perspective on mega-event economics. This time frame captures significant evolution in event scale, media commercialization, and global economic conditions, enabling analysis of how the cost-revenue relationship has changed over time.

DATA COLLECTION

For this project, the data utilized primarily came from the source Growth Dataset for Olympic Games and Football World Cups, publicly available through the Harvard Dataverse academic repository. This dataset was selected due to its comprehensive and well-documented collection of financial and economic data correlating to the Winter Olympics, Summer Olympics, and FIFA World Cup, dating from the 1960s to the late 2010s. This data set pulls information from multiple reliable sources, including official

Organizing Committee financial reports, government audits, International Olympic Committee (IOC) documentation, and FIFA financial statements. Supplemental data originates from peer-reviewed academic studies, economic analyses, and post-event audit summaries conducted by host countries. Other than this, data were sourced from local information posted in peer-reviewed online papers on economic data and on other large-scale sporting events, including the same event types the dataset covers for the missing years.

Within the dataset, several fields relevant to understanding the economic dynamics of hosting one of these large sporting events are contained. These include total event costs, venue and infrastructure spending, operational expenses, total revenue, sponsorship and broadcasting income, GDP trends surrounding the event year, and other socio-economic indicators. Because financial reporting practices vary across events and countries, the dataset compiles information from diverse formats and sources, standardizing them into a single structured file.

For the peer-reviewed papers, the same relevant information was found using web scraping methods. While the information gathered from these sources is not as in-depth as the dataset, similar fields like total event cost and total revenue were found and utilized.

For implementation into this project, the dataset was obtained in an Excel format and imported into Python, along with utilizing web-scraping tools supported by Python for the peer-reviewed papers for preprocessing and analysis. This format allowed for straightforward integration into data analysis tools such as Pandas and NumPy.

The dataset served as the central foundation for all subsequent work, including data cleaning, exploratory analysis, and model development. By relying on a publicly accessible, academically vetted dataset and peer-reviewed papers, this project ensures transparency, reproducibility, and consistency with economic literature examining the financial impact of global sporting events.

DATA CLEANING

Significant cleaning of data was required to make sure that the data was consistent and to prepare the data for analysis. There were major steps included, such as currency conversions, which were performed to standardize all the financial values into a unified currency, which would make it suitable for comparisons. Then the inconsistencies related to GDP and tourism metrics were addressed, since the dataset included missing entries, duplicates, and values in incorrect fields; the GDP values had to be standardized for easier comparison. Ensuring uniformity was there throughout the dataset was also important, making sure all

the data was in terms of numeric values to allow analysis and model training.

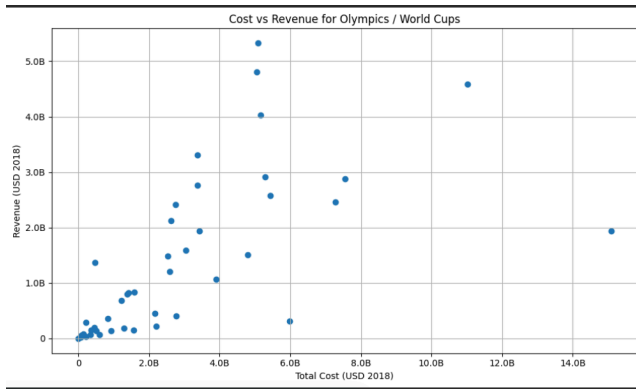
After cleaning, the dataset contained and displayed many variables that are essential for exploratory data analysis and model training. They include: Number of events, Number of countries, Revenues (ticketing, Broadcasting, International and Domestic Sponsorships), total cost, and GDP changes. By completing the steps, the dataset became reliable and consistent, which enabled meaningful data analysis and accurate model training in the following stages.

	Number of events	Number of countries	Ticketing Revenue (USD2018)	Broadcast Revenue (USD2018)	International Sponsorship Revenue	International Sponsorship Revenue (USD 2018)	Domestic Sponsorship Revenue (USD 2018)	Cost of venues (USD 2018)
count	29.0	29.0	29.0	29.0	24.0	29.0	0.0	29.0
mean	149.24	106.69	201.3M	892.9M	199.7M	232.5M	NaN	1.8B
std	102.18	60.31	254.6M	852.3M	211.4M	231.7M	NaN	2.7B
min	34.00	35.00	6.4M	7.3M	1.0M	846208.00	NaN	40.2M
25%	57.00	64.00	44.0M	108.1M	25.4M	28.3M	NaN	280.3M
50%	102.00	88.00	110.9M	693.7M	118.9M	157.4M	NaN	791.2M
75%	237.00	159.00	255.5M	1.4B	320.8M	359.2M	NaN	2.1B
max	306.00	207.00	1.1B	3.0B	668.0M	698.9M	NaN	12.6B

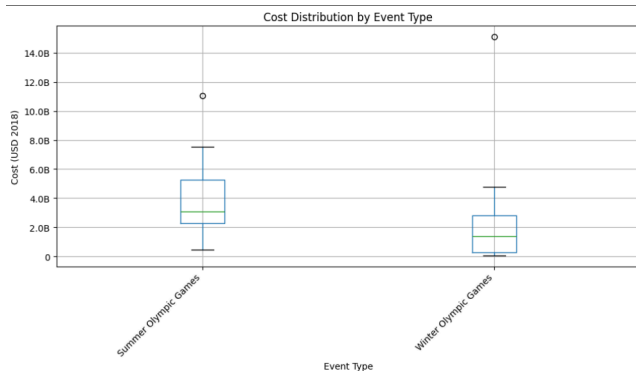
EXPLORATORY DATA ANALYSIS

Our exploratory data analysis revealed fundamental patterns and relationships in the historical mega-event dataset, providing crucial insights that informed subsequent modeling decisions and interpretation of results. Descriptive statistics quantified the central tendencies, variability, and distributions of key variables across the 56-year study period.

Analysis of total costs demonstrated substantial variation across events, with mean costs of approximately 4.2 billion USD (in 2018 values) and a standard deviation of 2.8 billion USD. The cost distribution exhibited right skewness, indicating that while most events clustered around moderate cost levels, several exceptionally expensive events pulled the mean higher than the median. Minimum costs approached 500 million USD for earlier, smaller-scale events, while maximum costs exceeded 15 billion USD for recent large-scale productions. This wide range reflects both trends toward increasing expenditure and event-specific factors such as existing infrastructure availability.

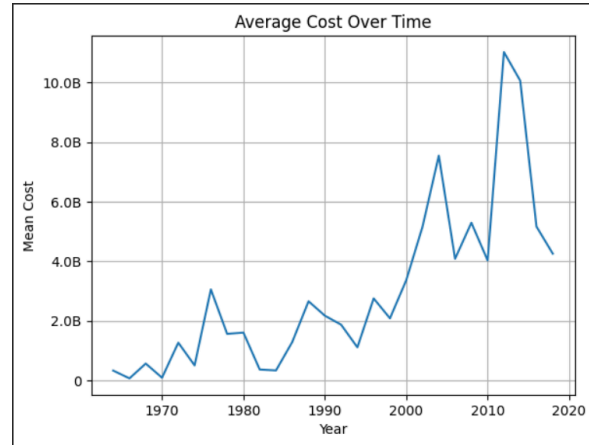


Revenue statistics revealed narrower distributions with a mean total revenue of approximately 1.6 billion USD and standard deviation of 800 million USD. Unlike costs, revenues showed less dramatic an increase over time, with the distribution more tightly clustered around the mean. Maximum revenues reached approximately 3.5 billion USD for the most commercially successful events, while the minimum revenues were several hundred million USD. The ratio of standard deviation to mean was substantially lower for revenues than for costs, indicating more predictable revenue patterns compared to highly variable cost structures.



A time-series visualization of total Olympic hosting costs further reinforces these statistical observations by showing a clear upward trajectory across the six decades in average cost. The cost-over-time graph revealed a pronounced long-term growth trend, with hosting expenses rising steadily from under 1 billion USD for events in the 1960s and early 1970s to well over 10 billion USD for recent Olympic Games. This escalation was not smooth. Instead, the graph exhibited several jumps relating to particularly infrastructure-heavy events like the 2008 Beijing Olympics and the 2014 Sochi Winter Games. Both of these events well exceeded historical spending levels. The increasing slope of the trend line lends itself to the fact that the financial burden of hosting continues to intensify at an accelerating rate. This is due in part to expanding venue requirements, increased security demands, growing

technological expectations, and urban development projects often tied to the Games.



MODEL TRAINING

The primary objective of the modeling process in this study was to determine whether financial and economic outcomes for large sporting events could be accurately predicted using historical data, and to evaluate whether future Olympic and World Cup events are likely to be profitable or cost the host countries financially. To accomplish this, we developed and evaluated multiple machine learning models using the cleaned dataset from the Harvard Dataverse. The modeling process followed a structured workflow: First, feature selection and data preparation. Followed by model construction, training, validation, model comparison, and, lastly, application of the model to predict future events and their costs.

For the first task, total revenue (based on the USD 2018) served as the dependent variable. Total event cost was used as the primary predictor. This predictor included infrastructure spending, venue construction, and organizational expenses. This demonstrates the assumption made by the group that higher spending on the event corresponds to greater expected revenue generated through broadcasting rights, ticketing, tourism, and sponsorships. Although the dataset includes secondary variables such as GDP changes and tourism indicators, these fields were not consistent across years. Due to this, these inconsistent fields were excluded to avoid any further biasing due to missing data points.

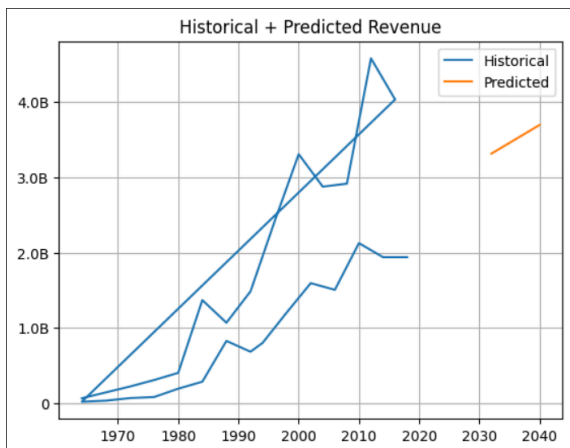
For the second task, the event year served as the independent variable, while the total cost of hosting the event was used as the output. This allowed for the modeling

of cost inflation across decades to be able to reliably predict the revenue and cost of future events. By combining the results of both models, we could generate an estimated profit gap for future events.

The first model implemented was a Linear Regression model, chosen for its interpretability and its common use in baseline economic forecasting. The model assumes a linear relationship between event cost and revenue. Training was conducted by splitting the historical dataset into an 80% training set and a 20% testing set. The model was then fit using least squares and minimizing the residual error between predicted and actual revenue.

Once trained, the model was evaluated using two primary performance metrics: R^2 (coefficient of determination) and Mean Absolute Error (MAE). The R^2 score for the linear regression model was -0.41 , indicating a very weak and potentially misleading relationship. A negative R^2 suggests that the linear model performed worse than simply predicting the mean revenue for all events. The MAE was approximately \$1.26 billion, meaning the model's predictions were, on average, off by more than a billion dollars. Such a high error reinforces the conclusion that revenue is not linearly dependent on cost for mega-events, likely because financial outcomes depend on numerous external factors. Although linear regression performed poorly, it served as a necessary baseline model.

```
R2: -0.4116793887486512
MAE: 1263773788.641535
Slope: 0.3616527105685546
Intercept: 197409725.77811646
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The second model used was a Random Forest Regressor, which is a method capable of modeling nonlinear relationships. Random Forests operate by constructing many decision trees, each trained on a random subset of the data, and averaging their predictions to lower the

chances of overfitting. This property makes them particularly effective in datasets with high variance or irregular patterns, which are present in our data.

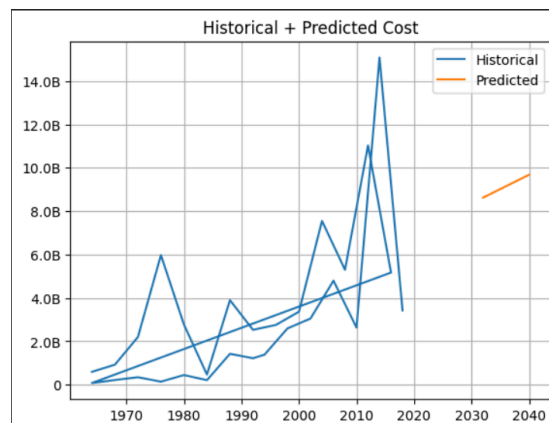
The Random Forest model was trained using the same features and data split as the linear regression model. Its performance was better compared to the linear model. The R^2 score increased to 0.23, and the MAE dropped to approximately \$1.04 billion. Although the error remained relatively high, the improvement across both metrics demonstrated that the Random Forests can capture the underlying structure of the data more accurately than linear regression.

```
Linear Regression Performance:
R2: -0.4116793887486512
MAE: 1263773788.641535

Random Forest Regressor Performance:
R2: 0.2342037269457573
MAE: 1040762171.5281504

Selected model for future predictions: RandomForestRegressor
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For future predictions cost for future events was needed. This was achieved through a separate Year Cost linear regression model, trained to estimate how event costs change over time. Unlike revenue, historical event costs show a clear upward trajectory. The model predicted that the cost in 2032 to be 8.6 billion USD, 9.2 billion USD in 2036, and 9.7 billion USD in 2040.



CONCLUSION

This research applied comprehensive data science methodologies to analyze the economic impact of major sporting events over a 56-year period, addressing the fundamental question of whether these mega-events deliver positive financial outcomes for host nations. Through systematic data collection, rigorous preprocessing, exploratory analysis, and predictive modeling using Linear Regression and Random Forest algorithms, the study

generated empirical evidence regarding cost-revenue patterns and future financial trajectories.

The primary conclusion is clear: major sporting events have consistently generated financial losses for host nations, with the magnitude of these losses increasing over time. Historical analysis reveals that the costs systematically exceed revenues, with break-even outcomes representing rare exceptions rather than typical results. The cost-to-revenue ratio demonstrates persistent values above 1.0, indicating structural financial challenges inherent in mega-event economics.

Future Projections based on the selected predictive models suggest deteriorating financial circumstances for events in 2032, 2036, and 2040. Predicted losses escalate from 4.2 billion USD to 5.1 billion USD over this period, driven by continued cost escalation to 6.6 billion USD while revenues stagnate around 1.5 billion USD. These projections assume continuation of historical trends, structural reforms in event planning and management could alter these trajectories, but absent such changes, the financial outlook remains challenging.

CCS CONCEPTS

• Computing methodologies -> Machine Learning • Applied Computing -> Economics • Information systems -> Data analytics

KEYWORDS

Gross Domestic Product (GDP), mega-events, major events, economic impact, data science, predictive modeling, model training, Olympic Games, FIFA World Cup, Linear Regression, Random Forest Model, Revenue, R^2 , MAE.

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